

(12) PATENT APPLICATION PUBLICATION

(21) Application No.202641027375 A

(19) INDIA

(22) Date of filing of Application :09/03/2026

(43) Publication Date : 20/03/2026

(54) Title of the invention : Software Quality Analysis and Risk Assessment Using Cloud-Based Machine Learning

(51) International classification	:G06F 11/36, G06N 20/00, G06F 9/44, G06Q 10/06, G06F 21/57	(71) Name of Applicant : 1)DIVA SENTHILNATHAN Address of Applicant :V.S.B. COLLEGE OF ENGINEERING TECHNICAL CAMPUS, NH-209, Coimbatore-Pollachi main road, Ealur Pirivu, Solavampalayam (po), Tamil Nadu India
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(86) International Application No	:	3)DR T KALAIKUMARAN
Filing Date	:01/01/1900	4)MRS V RADHA
(87) International Publication No	: NA	5)R.R.PRASANNARAM
(61) Patent of Addition to Application Number	:NA	
Filing Date	:NA	
(62) Divisional to Application Number	:NA	
Filing Date	:NA	

(57) Abstract :

Software quality has a critical role to play in the success of present-day applications. However, many organizations still rely on manual code reviews, static testing tools, and quality-checking measures after software development to detect defects. Their results are often obtained at later stages of the development process, which increases the cost of the project and delays the delivery date. This project suggests a cloud-based machine learning system which can be used for assessing software quality and determining the risks involved in the software development process. Instead of waiting for failures to manifest, the system predicts defect modules, security defects, and performance risks in the early stage. Such a risk ranking system is aimed at being based on the analysis of the source code metrics, historical defect records, test coverage reports, and development activity logs. The solution is deployed on a cloud platform, which enables scalable processing, data storage in one place, and easy integration with development pipelines. A web dashboard delivers clear visual insights that give developers, testers, and project managers access to faster and more informed decision-making. This transition from reactive test cases to risk prediction results in fewer failures after release, better maintainability, and enhanced overall reliability.

No. of Pages : 8 No. of Claims : 10